

SANKET MUCHHALA

AI /ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building production-grade AI systems using generative AI, NLP, and machine learning. Skilled in agentic AI, LLM pipelines, and scalable APIs. Experienced in delivering AWS-based AI solutions and collaborating with stakeholders across insurance and analytics domains.

EDUCATION

Master of Science in Data Science | Indiana University Bloomington, USA | Aug 2022 – May 2024

Bachelor of Technology in Information Technology | Thakur College of Engineering and Technology, India | Aug 2018 – May 2022

TECHNICAL SKILLS

Programming Languages:	Python, SQL, R, JavaScript
AI/ML Frameworks & Tools:	Scikit-learn, TensorFlow, PyTorch, FastAPI, MLflow, SpaCy
Generative AI & LLMs:	GPT-4 (via OpenAI APIs), LangChain, RAG, Agentic AI tools, Vector DBs(FAISS)
NLP & Document Intelligence:	Named entity recognition (NER), text classification, summarization, sentiment analysis
Data Engineering & Storage:	Pandas, NumPy, PySpark, AWS (SQS, Step Functions), Azure Data Lake, Azure SQL
Databases:	PostgreSQL, MySQL, Snowflake, Teradata
Visualization & BI:	Tableau, Power BI, R Shiny

PROFESSIONAL EXPERIENCE

AI Engineer | Progressive Insurance, Remote, USA | May 2024 – Present

- Built NLP workflows for claims data intake using Python and Azure ML, streamlining processing tasks by 25%.
- Designed Agentic AI workflows for document processing and decision support, enabling multi-step reasoning across claims data pipelines.
- Deployed GPT-4 based document summarization tools, reducing review cycles for compliance and legal teams by 40%.
- Designed executive-facing Tableau dashboards for AI outputs, supporting underwriting and fraud detection strategy.
- Improved fraud detection precision by 22% through real-time anomaly models integrated with SQL-based data streams.
- Built AWS-based MLOps pipelines (Step Functions, SQS) for model monitoring, retraining, and deployment, reducing drift incidents by 35%.
- Developed FastAPI APIs to integrate LLM services into enterprise workflows, enabling deployment across claims and legal platforms.
- Partnered with business stakeholders (compliance, legal, underwriting) to define LLM explainability and deployment standards.
- Led cross-functional sessions to communicate AI system behavior and outputs, improving adoption and stakeholder trust by 35%.
- Implemented a governance framework for LLM usage with prompt logging, version control, and access auditing for audit readiness.

Research Assistant – Generative AI | Indiana University Bloomington, IN, USA | Dec 2023 – May 2024

- Improved transcript accuracy by 18pp using a GPT-4 RAG pipeline deployed on BigRed200, processing over 200 hours of esports video.
- Built and optimized **real-time LLM-based microservices**, reducing latency by 40% for large-scale chat systems.
- Automated retraining pipelines using SLURM on HPC systems, cutting manual ETL effort by 6 hours per match.
- Documented GenAI workflows, adopted by two graduate cohorts for ongoing esports psychology research.

Data Analyst | IBM, MH, India | Sep 2020 – Jun 2022

- Led end-to-end development of a churn prediction model using Python and Scikit-learn, driving a 20% reduction in customer attrition.
- Refactored ETL workflows using Azure Data Lake and SQL, improving data availability and cutting processing time by 15%.
- Built automated data validation pipelines with SQL and Python, raising dashboard reporting accuracy by 18%.
- Deployed ML models to Azure ML environments with CI/CD support, accelerating release cycles by 25% across internal products.
- Authored reproducible model documentation and Jupyter-based reports to align analytics delivery with stakeholder needs.
- Collaborated with stakeholders and solution architects to align ML models with business objectives and deployment constraints.
- Introduced versioning standards for ML pipelines and datasets, increasing transparency in model updates and audits.
- Conducted internal training sessions on Python-based analytics tooling, improving team adoption of reusable code modules.

PROJECTS

Nerdplexity: [Github](#)

Built end-to-end ML pipelines for production, improving model accuracy by 22% and reducing drift incidents by 35% through automation. Designed RAG-based AI systems using embeddings and vector search, cutting response time by 35% and boosting personalization accuracy by 30%.

kemlang.cloud: [Github](#)

Built KemLang, a custom programming language in Python with lexical analysis, parsing, AST generation, and interpretation.

Shipped a full developer experience for KemLang with CLI commands, web playground, VS Code syntax support, and test automation.

LexOrchestrator: [Github](#)

Built multi-agent legal AI workflows for citation validation, RAG retrieval, and hallucination reduction in document review pipelines v2. Designed orchestrator and sub-agent pipelines with eval loops and tool routing for high-reliability legal reasoning tasks production use.

Certifications:

AWS Certified MLE Associate, Azure AI Fundamentals(AI-900), Databricks Generative AI Foundations, Google Data Analytics